

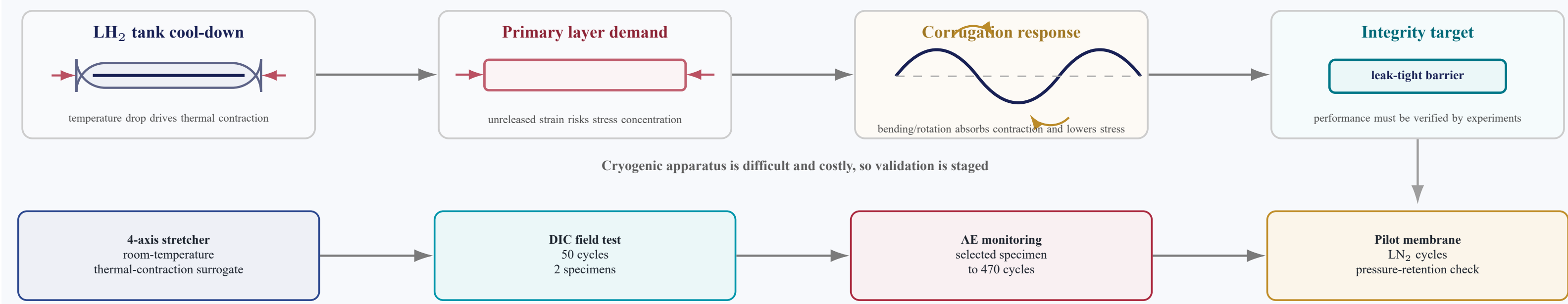
Introduction

Large-scale LH₂ cargo containment needs a metallic primary barrier that is leak-tight, formable, and tolerant of the in-plane displacement generated when the tank system cools from ambient temperature to cryogenic service. If this displacement is carried mainly by membrane stretching, local strain concentration can become a direct risk for cracking, leakage, and loss of containment reliability.

A corrugated 316L membrane is introduced to turn this constraint into a geometric compliance problem. The selected cross-corrugated profile is expected to accommodate imposed displacement through local bending and rotation of crests, valleys, and side-wall transitions, thereby lowering the uniform stretching demand on the thin sheet.

The remaining question is experimental rather than purely geometric: after patent-based forming, specimen preparation, and cyclic loading, does the membrane still follow the intended strain-release mechanism, and can an assembled module retain preliminary integrity after cryogenic cycling?

Because full cryogenic apparatus is difficult and costly, this poster uses a staged validation chain: a room-temperature four-axis stretching apparatus simulates the contraction demand, DIC maps global strain redistribution during the first 50 cycles of two specimens, AE follows damage activity for a selected specimen to 470 cycles, and a pilot membrane is checked by LN₂ cycling plus pressure-retention testing.



M. Methods: Design-to-Test Logic Chain

Starting point	Value	Why it matters
Selected profile	$c_1 = 1.0, c_2 = 5.0, \lambda = 0.651$	best performing geometry from the previous design screen
Sheet and material	316L, $t = 1.2$ mm	thin metallic primary-barrier candidate
Patent basis	CN118237450A	coupled downward pressing and lateral squeezing for equal-section cross corrugation

1. Profile
previous best

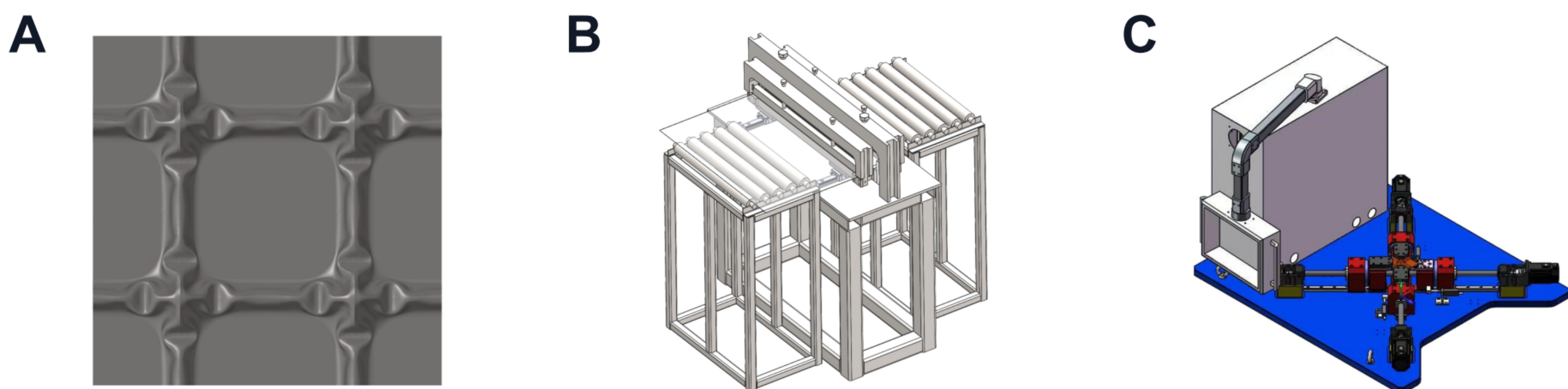
2. Patent process
die + side extrusion

3. Samples
form + cut

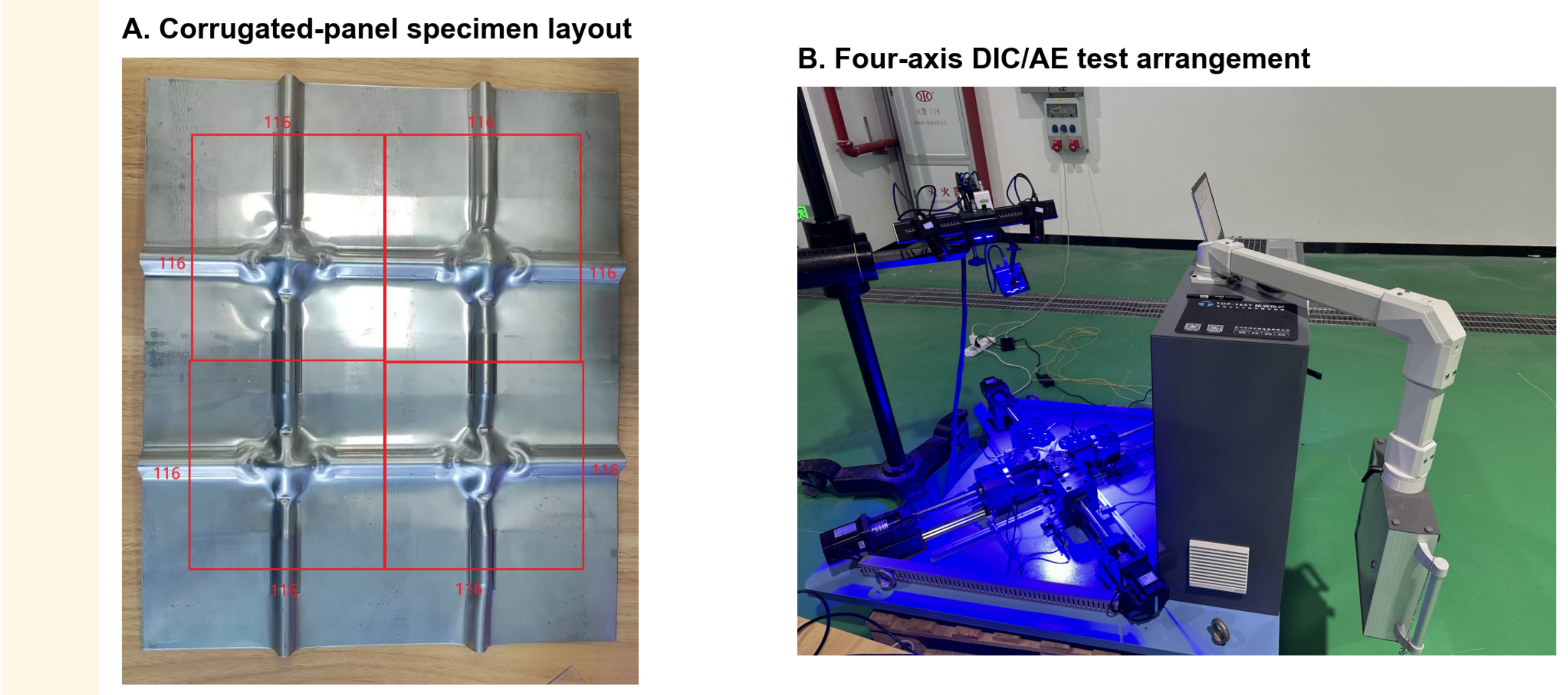
6. Assembly
LN₂ cycles

5. AE next
damage evolution

4. DIC first
strain release

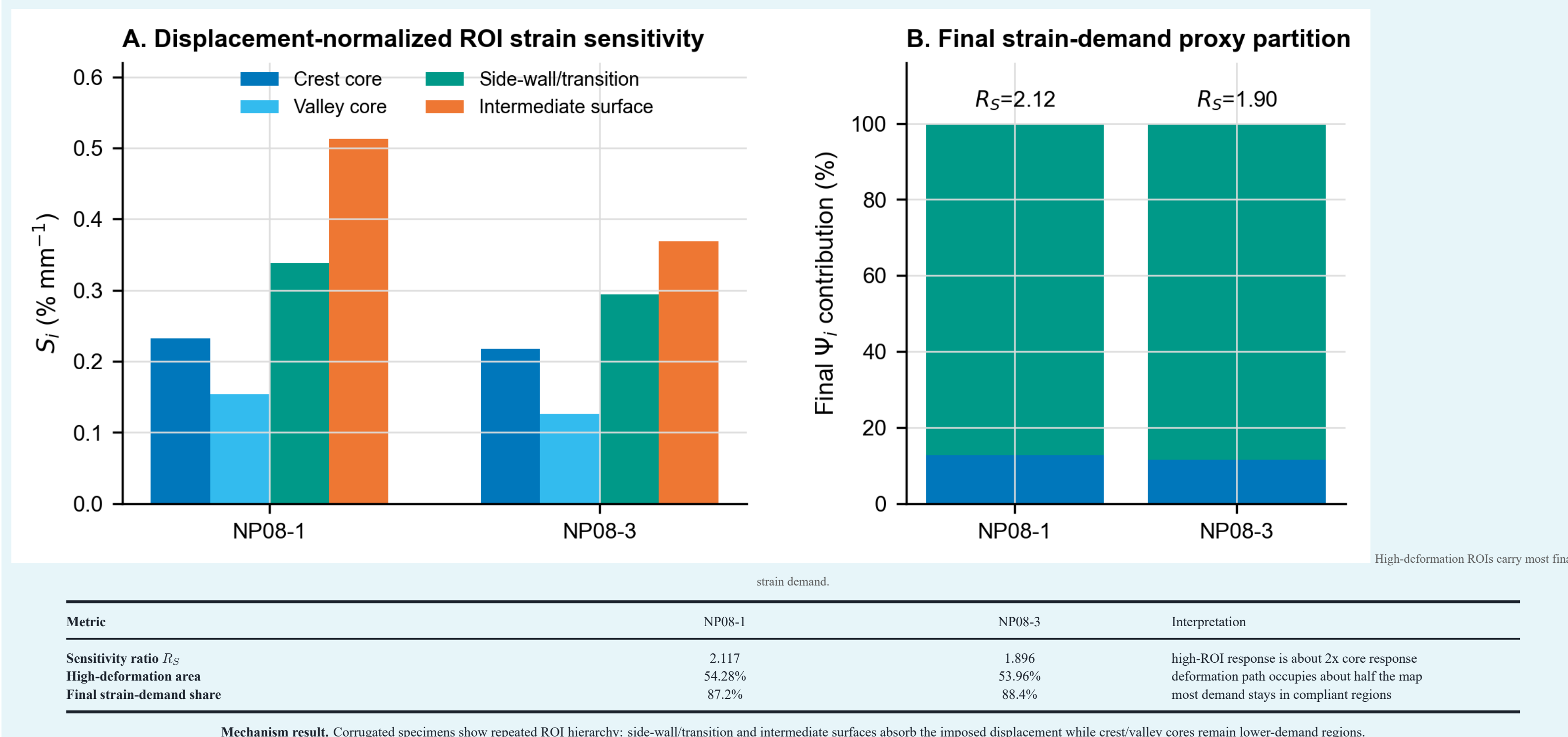
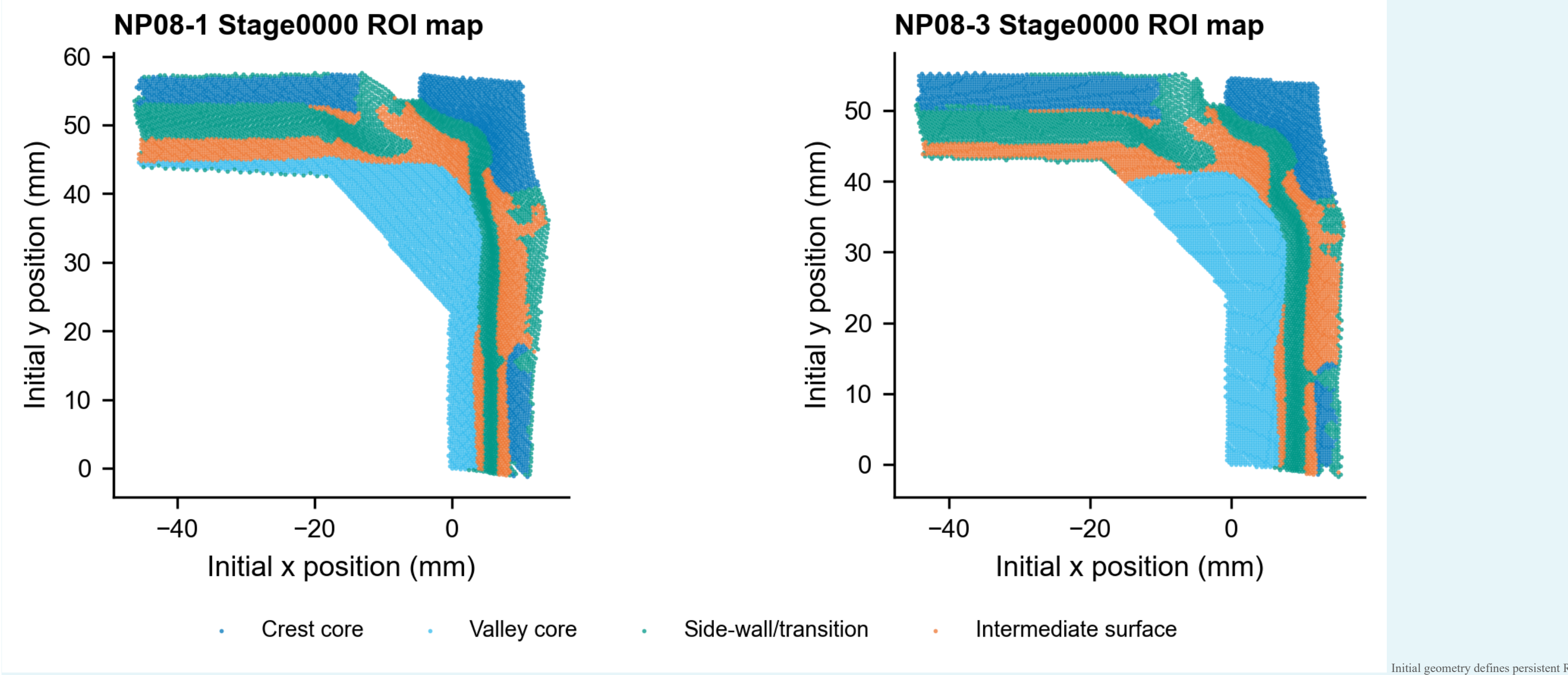
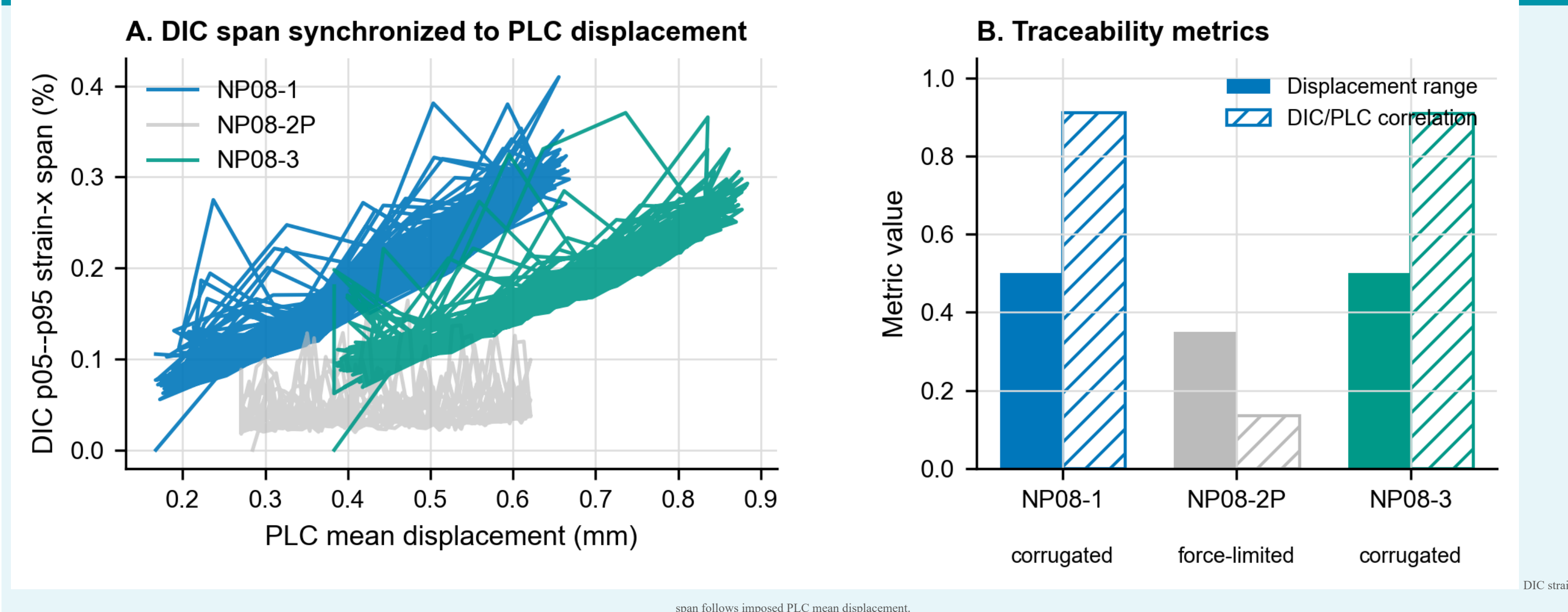


Selected profile → patented processing machine → four-axis specimen platform. The process figure is used as design and preparation context, not as a dimensional CAD claim.

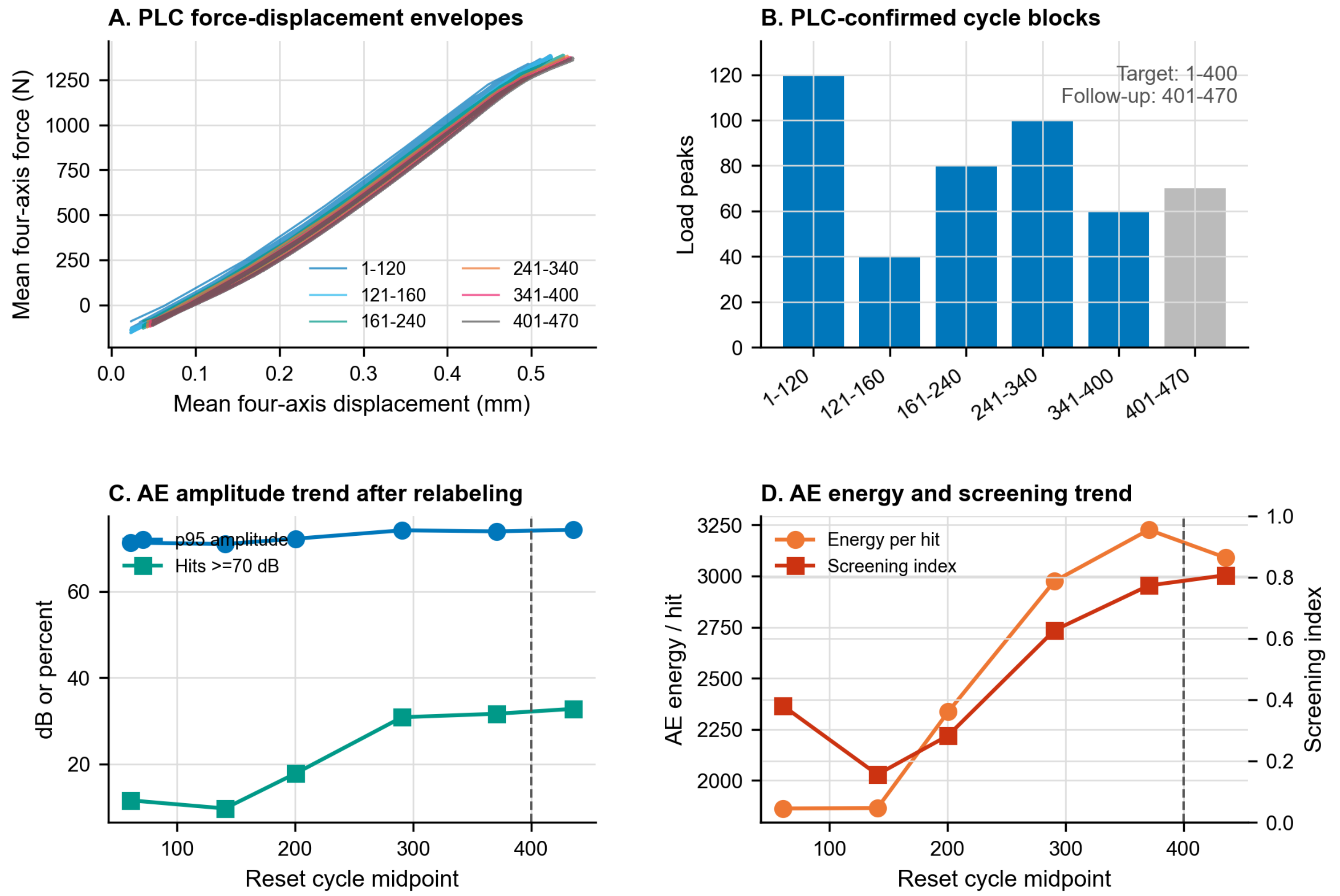


Experiment	Direct question	Measured output
DIC/PLC	How does corrugation release imposed strain?	synchronized strain span, geometry-fixed ROI hierarchy
AE	When does cyclic activity concentrate?	established cycle windows and damage-evolution screen
LN ₂ assembly	Does pressure-retention testing verify assembly integrity after thermal cycling?	pressure-hold and helium leak checks across five LN ₂ cycles

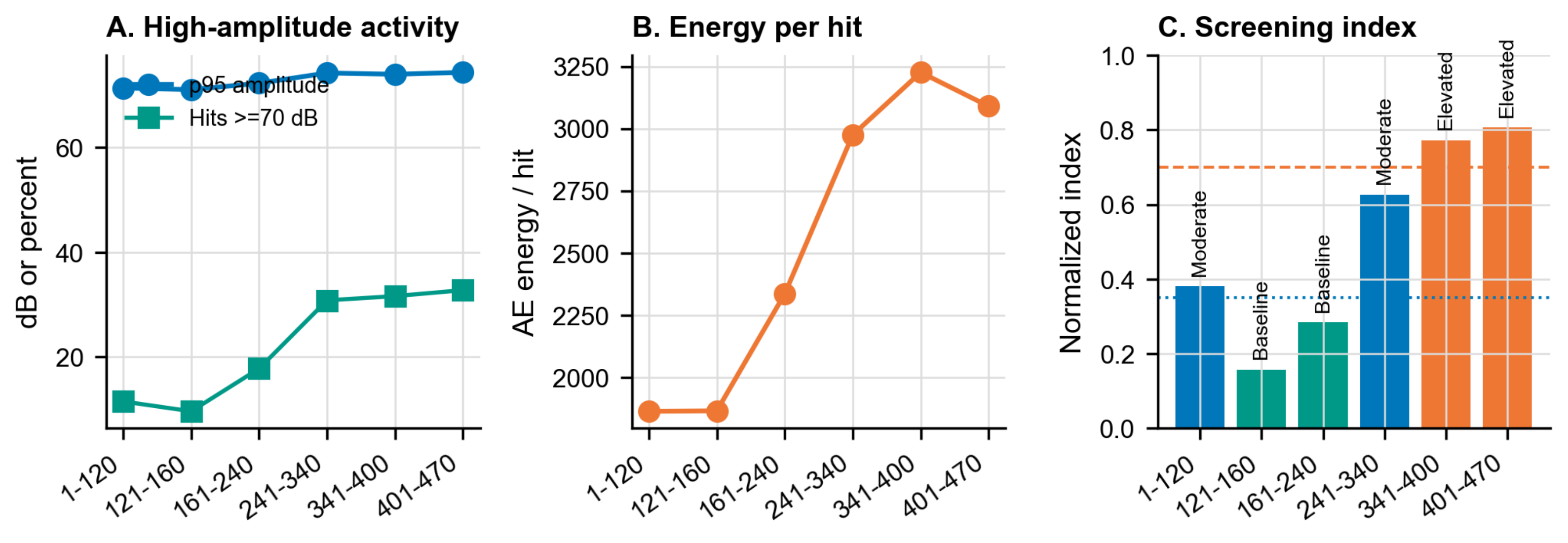
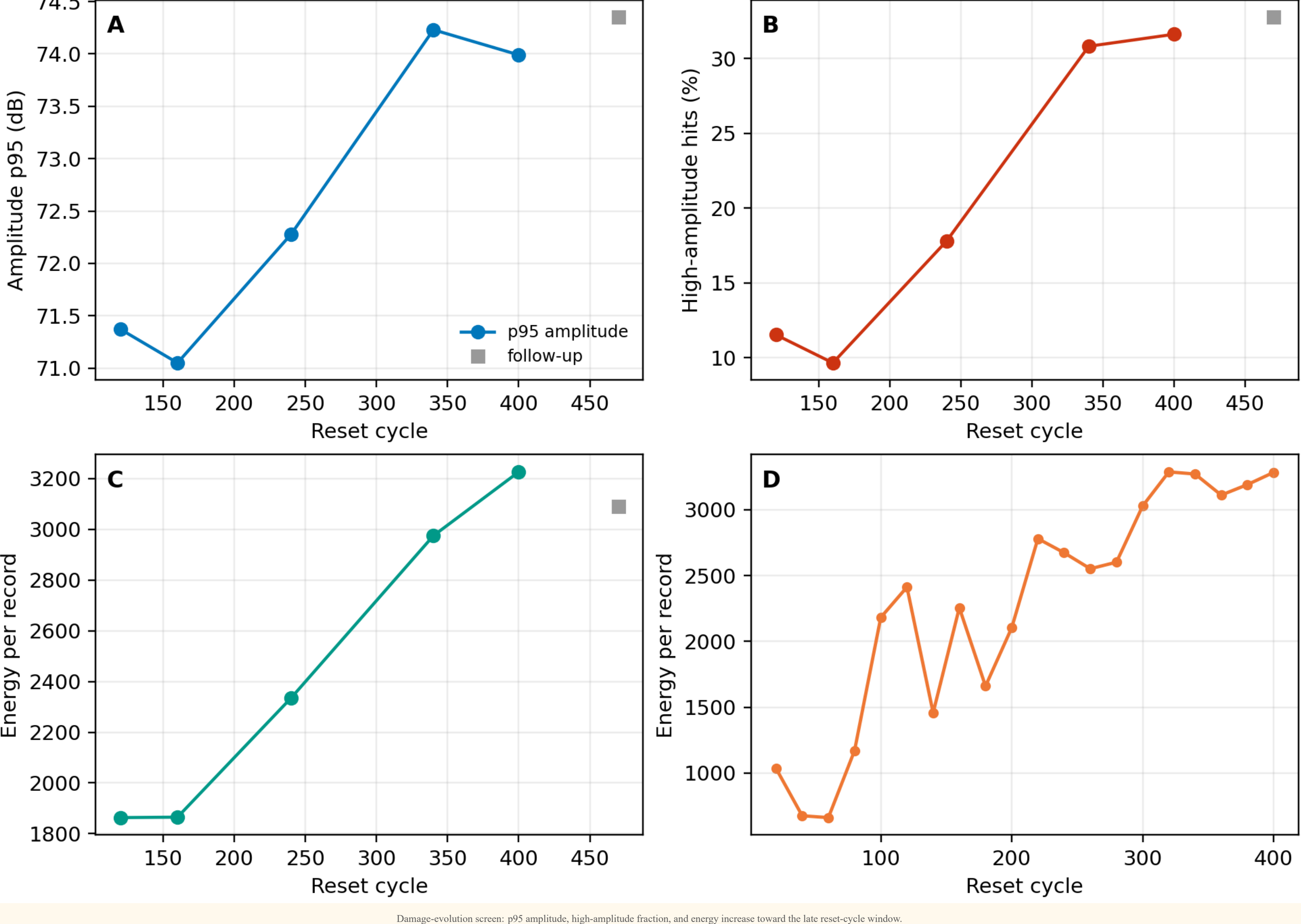
R1. Results: DIC/PLC Mechanism Evidence



R2. Results: AE Activity and Assembly Integrity Checks



Feature-level AE diagnostics after PLC cycle relabeling



31.63%
records above 70 dB

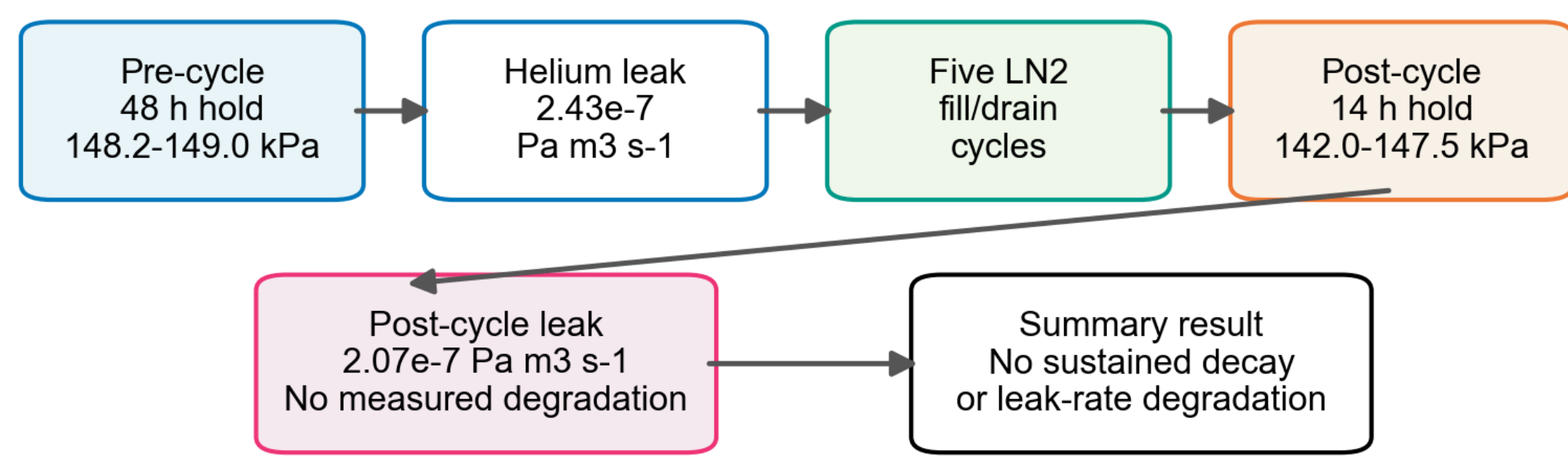
3.23×10^3
energy per hit

125 kHz
median frequency

1.17M
feature records

Scaled-prototype LN₂ assembly-integrity check

Author-held project summary: 0.2 m3 integrated membrane, pressure-retention testing, helium leak detection, and five LN₂ filling/draining cycles.



Claim boundary: pressure-retention is the test method; the supported objective is preliminary assembly integrity, not LH₂ qualification.

D. Discussion: What the Evidence Supports

Layer	Supports	Does not claim	Evidence panel
DIC/PLC	corrugation-driven strain redistribution	service lifetime	R1
AE	late-cycle inspection-priority window	crack growth proof	R2-AE
LN ₂	preliminary assembly integrity	LH ₂ qualification	R2-LN ₂

The strongest evidence is mechanism evidence from synchronized DIC/PLC and ROI mapping.
AE is useful after relabeling because it ranks when inspection attention should focus.
LN₂ pressure-retention testing is necessary but not sufficient for final LH₂ service readiness.

C. Conclusions and Next Gates

Conclusions

- Corrugation redistributes strain into traceable geometry-defined ROIs.
- The late AE reset window gives an inspection-priority signal.
- Pre/post LN₂ pressure and leak checks support preliminary assembly integrity.

Next gates

- Repeat coupon tests with ROI uncertainty.
- Add cryogenic DIC and waveform-level AE.
- Report full acceptance pressure/leak records under LH₂-relevant criteria.